SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	K. Ragavi	**Reg. No.:**	192524364
Section / Batch:	B. Tech AI&DS/2ND	**Date:**	13/07/2026

Code of Conduct

I K. Ragavi/192524364 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Ragavi

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
		GRAND TOTAL:	**100 Marks**

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

SMART QUEUE MANAGEMENT FOR CAFETERIA TOKEN SYSTEMS USING FIFO DATA STRUCTURE

Presented by
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Guided by
Dr Uma Priyadharshini

Introduction

- A cafeteria token system is used to manage customer service in an organized manner.
- During peak hours, a large number of customers arrive simultaneously.
- Manual token management often results in delays and confusion.
- An efficient data structure is required to maintain the order of service.
- The Queue Abstract Data Type (ADT) provides an effective solution for this problem.

Queue ADT and FIFO Principle

- A Queue is a linear data structure that follows the First In, First Out (FIFO) principle.
- The first customer to receive a token is the first customer to be served.
- New customers are added to the rear of the queue using the enqueue operation.
- Customers who have been served are removed from the front using the dequeue operation.
- The FIFO principle guarantees fair and systematic customer service.

Smart Queue Management System

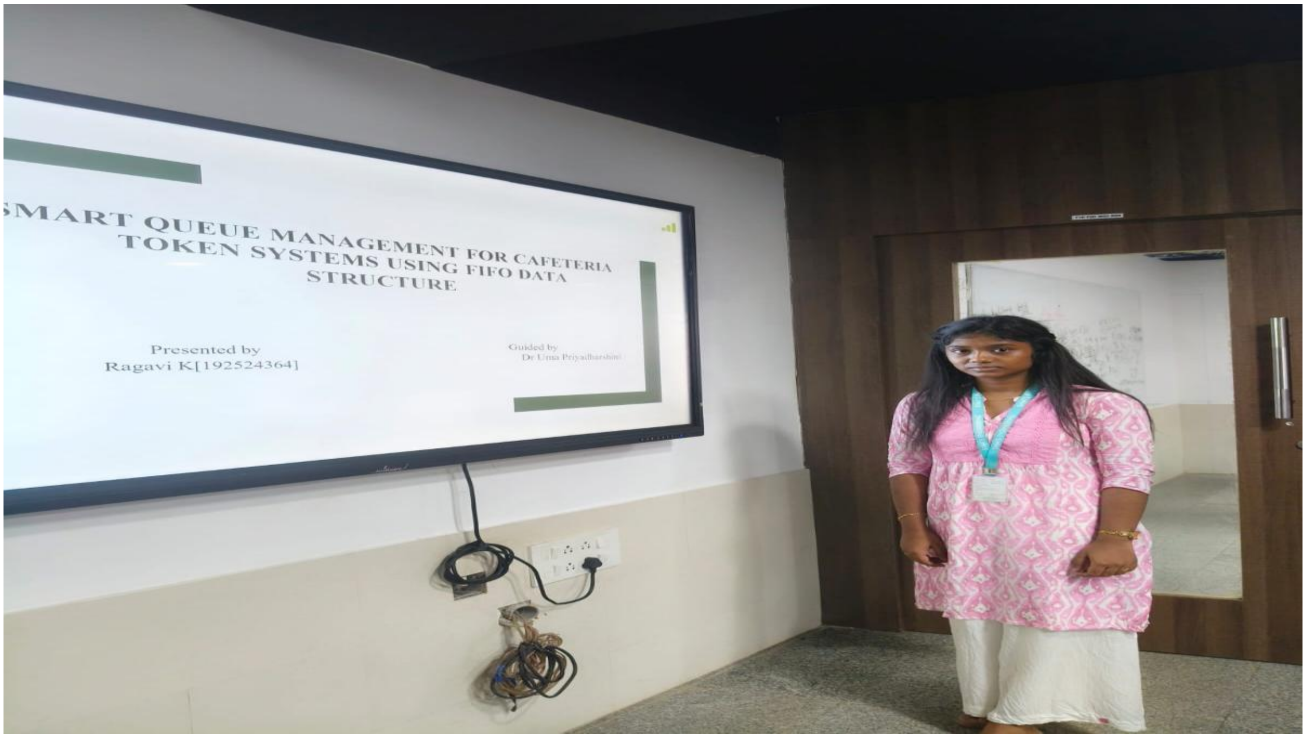
- Each customer is assigned a unique token upon arrival.
- The issued token is added to the rear of the queue.
- Customers are served strictly according to their token sequence.
- After service, the corresponding token is removed from the front of the queue.
- The system continuously updates the queue to maintain an accurate waiting list.

Performance Analysis

- The enqueue operation inserts a new token into the queue in **O(1)** time.
- The dequeue operation removes the front token in **O(1)** time.
- The Queue ADT minimizes customer waiting time through efficient processing.
- The system maintains service order without unnecessary delays.
- Queue-based management improves the overall efficiency of cafeteria operations.

Advantages and Conclusion

- The Queue ADT ensures fair customer service by following the FIFO principle.
- It provides a simple, reliable, and efficient solution for token management.
- The system reduces crowding and eliminates service-order confusion.
- Queue-based token management enhances customer satisfaction and operational efficiency.
- Therefore, the Queue ADT is the most suitable data structure for cafeteria token systems.



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____________	Signature: _____________	Signature: _____________
Date: _________________	Date: _________________	Date: _________________